

Def.

A subset $S \subseteq \mathbb{R}^3$ is called a Regular Surface if:

for each $p \in S$, there are open neighbor $V \subset \mathbb{R}^3$ containing p and open $U \subset \mathbb{R}^2$ with

$X: U \rightarrow V \cap S$ which satisfies

- 1). X is C^∞ .
- 2). X is Homeomorphism
- 3). dX_q is 1-1 for each $q \in \mathbb{R}^2$.

Ellipsoid.

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\}, \quad (a, b, c \neq 0)$$

Define $f: \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \mapsto \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$

Then, $E = f^{-1}[\{0\}]$, f is differentiable because it is polynomial
and $0 \in f[E]$ is a regular point, because $f(x, y, z) = 0$ iff $(x, y, z) = \left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right)$

but at this point, $df = (f_x \ f_y \ f_z) \neq 0$, so df is surjective.

Now, by the theorem, E is regular surface

Torus.

Consider a circle S' centered at $(0, a, 0)$, with a equation

$$(y-a)^2 + z^2 = r^2.$$

Rotate this with respect to z -axis, then we obtain a torus

$$z^2 = r^2 - (\sqrt{x^2 + y^2} - a)^2$$

That is, the torus is, for a map

$$f(x, y, z) = z^2 + (\sqrt{x^2 + y^2} - a)^2,$$

$$T = f^{-1}[\{r^2\}].$$

f is differentiable excepte $(x, y) \neq (0, 0)$, with

$$\frac{\partial f}{\partial x} = 2 \cdot (\sqrt{x^2 + y^2} - a)^2 \cdot \frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial f}{\partial y} = 2 \cdot (\sqrt{x^2 + y^2} - a)^2 \cdot \frac{y}{\sqrt{x^2 + y^2}},$$

$$\frac{\partial f}{\partial z} = 2z$$